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SpeakX

Project Aurora

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## Introduction

Across EdTech, Consumer Apps, FinTech, HealthTech, and SaaS—user communication is the single most powerful, yet most poorly engineered, growth lever. Today's notification systems are rule-based, segment-blind, timing-agnostic, and completely unaware of what actually works for whom, when, and why.

By 2030, intelligent communication orchestration will be the competitive moat—knowing WHO to talk to, WHAT to say, WHEN to say it, WHY it works, and HOW to improve continuously. Your challenge is to build that system: a self-learning notification orchestrator that can extend to any B2C or B2B application.

EdTech (SpeakX) is the reference domain, but your architecture must be domain-agnostic—swap the Knowledge Bank and user data, keep the same orchestrator core.

## About Us

SpeakX is an AI-powered English speaking and communication learning app that helps Hindi and regional language speakers in India transform from "I understand English but can't speak" to confident English speakers through daily, guided speaking practice with our AI teacher "Sia".

We target pan-India across Tier 1, 2, and 3 cities, with Tier 2 and Tier 3 being our primary audience. Our core users are working professionals, college students, and ambitious homemakers in the 20-45 age group seeking better jobs, promotions, and social confidence.
With features like AI tutor conversations, gamification (coins, streaks, leaderboards), progress reports, and role-based practice scenarios, we drive daily engagement and habit formation.

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Key metrics include Trial→Monthly conversion, Monthly Retention (Users who complete an exercise within the Month out of all users who converted from trial to monthly user), W1 Retention (Users who complete an exercise within the 1st week after trial to monthly conversion out of all users who converted from trial to monthly user) across different user personas. These numbers represent significant opportunities for improvement through intelligent, personalized communication.

## Problem Statement Description

Design and build a fully working, self-learning notification and communication orchestrator that demonstrates how user engagement can become smarter, more personalized, and continuously improving. Your solution should combine data intelligence, AI-powered content generation, and behavioral learning across three integrated tasks:

## Task 1: System Architecture & Intelligence Design

Build the core intelligence engines that power the orchestrator:

- Knowledge Bank Engine: Ingest company KB (text/markdown/PDF), auto-infer North Star metric, extract feature-goal mappings, define allowed tones and Octolysis-inspired hooks
- User Data Ingestion: Accept behavioral CSV data (demographics, psychographics, feature usage, temporal patterns), validate schema, handle missing data, create user profiles
- MECE Segmentation Engine: Generate 6-12 mutually exclusive segments with propensity scores (gamification, AI tutor, leaderboard, social), activeness scores, and churn risk signals
- Goal & Journey Builder: Define primary goals, sub-goals, and day-on-day progression for each segment × lifecycle stage (Trial D0-D7, Paid D8-D30, Churned/Inactive)

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## Task 2: Communication & Timing Intelligence

Build the messaging and timing optimization layers:

- Theme Engine: Define most relevant communication themes per segment using Octolysis 8 Core Drives (Epic Meaning, Accomplishment, Empowerment, Ownership, Social Influence, Scarcity, Unpredictability, Loss Avoidance)
- Template Generator: Create exactly 5 message templates per Segment × Lifecycle × Goal × Theme combination, with bilingual content (Hindi + English), tone, hook, and feature reference
- Timing Optimizer: Define 6 standard time windows (early_morning 06-09, mid_morning 09-12, afternoon 12-15, late_afternoon 15-18, evening 18-21, night 21-24), learn optimal windows per segment
- Frequency Optimizer: Determine 3-9 notifications/day based on segment activeness (>0.7 = 7-9/day, 0.4-0.7 = 5-6/day, <0.4 = 3-4/day) with guardrail override (uninstall >2% = reduce by 2)

## Task 3: Execution & Self-Learning

Build the output generation and learning feedback loop:

- Schedule Generator: Output user-wise notification schedule (user_id, segment, lifecycle_day, notif_1 through notif_9 with template_id, time, channel) showing gradual journey progression
- Classification: When performance data (On Demo day, experiment_results provided by SpeakX) is uploaded, classify templates as GOOD (CTR>15%, Engagement>40%), NEUTRAL (5-15%, 20-40%), or BAD (<5%, <20%)
- Learning Engine: Suppress BAD templates, promote GOOD as references, learn timing patterns per segment, generate improved Iteration 1 outputs with measurable delta
- Delta Reporter: Document every change (entity_type, entity_id, change_type, metric_trigger, before_value, after_value, explanation) with causal reasoning

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## Input Data Schemas

| Column | Type | Description |
|---|---|---|
| user_id | string | Unique anonymized user identifier |
| lifecycle_stage | enum | trial \| paid \| churned \| inactive |
| days_since_signup | int | Days since registration |
| age_band, region | string | Demographics (18-24/25-34/35-44/45+, tier1/tier2/tier3) |
| sessions_last_7d, exercises_completed_7d | int | Behavioral activity counts |
| streak_current, coins_balance | int | Gamification state |
| feature_ai_tutor_used, feature_leaderboard_viewed, | bool | Feature usage flags |
| preferred_hour | int | Most active hour (0-23) |
| notif_open_rate_30d, motivation_score | float | Engagement metrics (0-1) |

## Octolysis 8 Core Drives Reference

| # | Core Drive | Example Message Hook |
|---|---|---|
| 1 | Epic Meaning | "Join 1M+ learners transforming their careers" |
| 2 | Accomplishment | "You completed 5 lessons! Keep the momentum" |
| 3 | Empowerment | "Choose your learning path today" |
| 4 | Ownership | "Your 50 coins are waiting to be spent" |
| 5 | Social Influence | "3 friends joined this week. Invite more!" |
| 6 | Scarcity | "Only 2 days left in your trial!" |
| 7 | Unpredictability | "Surprise reward inside today's lesson!" |
| 8 | Loss Avoidance | "Your 7-day streak is at risk!" |

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## Experiment Results CSV Schema (Provided for Demo2 by SpeakX)

| Column | Type | Description |
|---|---|---|
| template_id | string | Unique template identifier |
| segment_id | string | Target segment ID |
| lifecycle_stage | string | User lifecycle stage |
| goal | string | Template goal alignment |
| theme | string | Octalysis hook used |
| notification_window | string | Time window of delivery |
| total_sends | integer | Number of notifications sent |
| total_opens | integer | Number of notifications opened |
| total_engagements | integer | Number of in-app actions taken |
| ctr | float | Click-through rate (opens / sends) |
| engagement_rate | float | Engagement rate (engagements / opens) |
| uninstall_rate | float | Uninstall rate post-notification |
| performance_status | string | Template classification (GOOD/BAD/ NEUTRAL) |

## Default Thresholds & Configuration

### Template Classification Cutoffs

| Classification | CTR | Engagement Rate | Action |
|---|---|---|---|
| **Good** | > 15% | > 40% | Promote, use as reference |
| **Neutral** | 5-15% | 20-40% | Iterate, A/B test further |
| **Bad** | < 5% | < 20% | Suppress, do not use |

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## Notification Frequency Bands

| Activeness Score | Notifs/day | Churn Risk | Strategy |
|---|---|---|---|
| > 0.7 (Highly Active) | 7-9 | Low | Maximum engagement |
| 0.4-0.7 (Moderate) | 5-6 | Medium | Balanced approach |
| < 0.4 (Low Active) | 3-4 | High | Conservative, avoid fatigu |

GUARDRAIL OVERRIDE: If segment uninstall_rate > 2%, reduce notification frequency by 2/day regardless of activeness score.

## Standard Time Windows

| Window Name | Time Range | Typical Use |
|---|---|---|
| early_morning | 06:00 - 08:59 | Morning motivation, habit trigger |
| mid_morning | 09:00 - 11:59 | Work break reminder |
| afternoon | 12:00 - 14:59 | Lunch break engagement |
| late_afternoon | 15:00 - 17:59 | Productivity boost |
| evening | 18:00 - 20:59 | Post-work learning |
| night | 21:00 - 23:59 | End-of-day recap, streak save |

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## Deliverables

1. Task 1: System Architecture & Intelligence
   - company_north_star.json - Inferred North Star, justification, measurable proxy
   - feature_goal_map.json - Feature → lifecycle → outcome mapping
   - allowed_tone_hook_matrix.json - Allowed/disallowed tones, hook taxonomy
   - user_segments.csv - MECE segments with propensities, activeness, churn risk
   - segment_goals.csv - Goals, sub-goals, day-on-day focus per segment × lifecycle

2. Task 2: Communication & Timing Intelligence
   - communication_themes.csv - Primary/secondary themes, tone preferences, hooks per segment
   - message_templates.csv - 5 templates per combination with bilingual content (Hindi + English)
   - timing_recommendations.csv - Optimal windows per segment with expected CTR/engagement
   - user_notification_schedule.csv - User-wise daily schedule (3-9 notifs with template, time, channel)

3. Task 3: Execution & Self-Learning
   - experiment_results.csv - (Provided by SpeakX) Performance data with CTR, engagement per template/timing
   - learning_delta_report.csv - Changes between iterations with causal explanations
   - Demo showing both iterations (before/after learning) in action
   - Complete downloadable codebase (Private GitHub repository)
   - README.txt (≤500 words) detailing architecture, models, and run instructions

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## Live Demo Flow

Phase 1: Initial Run (Iteration 0 - Before Learning)

- Evaluator provides Knowledge Bank (company KB text/markdown)
- Evaluator provides User Behavioral Dataset (CSV per schema)
- System processes inputs and generates ALL Iteration 0 outputs
- Contestant exports files to iteration_0_before_learning/ folder
- Evaluator reviews: segmentation, templates, timing, frequency, schedules

Phase 2: Learning Run (Iteration 1 - After Learning)

- Evaluator classifies templates as GOOD/NEUTRAL/BAD per cutoffs
- Evaluator provides experiment_results.csv (CTR, engagement per template/timing)
- System ingests performance data
- System relearns and generates improved Iteration 1 outputs
- Contestant exports files to iteration_1_after_learning/ folder
- Evaluator compares Iteration 0 vs 1 for MEASURABLE improvement

## Non-Negotiable Constraints

- Fully working end-to-end system with complete downloadable codebase
- Runnable locally for live demos (to catch/prevent hacky workarounds)
- Accept NEW datasets at runtime (we provide data on-the-spot during finals)
- Demonstrate REAL learning (Iteration 0 → Iteration 1 with measurable delta)
- Export ALL outputs in strict CSV/JSON formats per schemas defined
- Be explainable, auditable, and reproducible

AUTO-FAIL: Hardcoded outputs • Mock learning • PPTs without exports • No delta between iterations • Missing/renamed files • Merged iterations

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## Submission Structure

```
SpeakX_AURORA_<TeamName>.zip
├── iteration_0_before_learning/
│   ├── company_north_star.json
│   ├── feature_goal_map.json
│   ├── allowed_tone_hook_matrix.json
│   ├── user_segments.csv
│   ├── segment_goals.csv
│   ├── communication_themes.csv
│   ├── message_templates.csv
│   ├── timing_recommendations.csv
│   └── user_notification_schedule.csv
├── iteration_1_after_learning/
│   ├── user_segments.csv (if changed)
│   ├── message_templates.csv (updated)
│   ├── timing_recommendations.csv (updated)
│   └── user_notification_schedule.csv (updated)
├── experiment_results.csv
├── learning_delta_report.csv
├── codebase/ (complete runnable code)
└── README.txt (≤500 words)
```

## Evaluation Criteria

| Dimension | Weight | Description |
|---|---|---|
| System Completeness | 15% | All files present, correct schemas, runnable codebase, proper structure |
| Segmentation Quality | 15% | MECE compliance, meaningful descriptions, differentiated propensities |
| Messaging Intelligence | 25% | Template variety, theme relevance, bilingual quality, hook diversity, gradual journey |
| Timing & Frequency | 10% | Segment-aware frequency (3-9/day), timing optimization, guardrail respect |
| Learning & Evolution | 25% | Observable delta, causal explanations, real improvement in Iteration 1 |
| Extensibility | 5% | Domain-agnostic architecture, configurable for other industries |
| Presentation | 5% | Overall clarity, professionalism, demo quality, README documentation |